

# Problem Statement Worksheet (Hypothesis Formation)

## Predicting Videos That Stay Trending: A Closer Look At YouTube's Frequently Trending Content

### 1 Context

YouTube updates its trending list every 15min using a combination of factors such as the number of views, temperature (i.e. how quickly the video generates views), age of video, etc. Although its not clear what factors keep a video popping up in the trending list repeatedly. Identifying such features in the data could bolster the success of a video (i.e. make it frequently trend).

### 2 Criteria for success

Develop a machine learning classifier to predicts whether a YouTube video will reappear in trending list. The model uses 1/more features like: Viewer count, Engagement metrics, Age of a video and Temperature. It should be completed within 2 weeks.

### 3 Scope of solution space

This analysis will primarily be restricted to the measurable parameters which is in the control of a YouTuber. Trending is selected based on appeal, relevance and content that capture the breadth of what's happening across the world. Thus, multi-country data is used for analysis (and not country specific).

### 4 Constraints within solution space

Data is available for many regions; thus, video title is not necessarily restricted to a single language.

### 5 Stakeholders to provide key insight

*YouTube Content creators & Marketing experts* can benefit the most from this analysis.

### 6 Key data sources

Kaggle: [Trending YouTube Video Statistics](#)